COMP3308/3608 Week 11 - Probabilistic Reasoning & Bayesian Networks

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1 Motivation and Setup

Classical logic-based AI struggles with uncertainty: incomplete information, noisy observations, and stochastic environments. Probability theory provides a rigorous formalism for representing and manipulating uncertain beliefs. The central computational task is **inference**: given partial observations (evidence), compute the posterior distribution over variables of interest.

2 Probability Foundations

2.1 Basic objects

Definition 2.1.1 (Random variable)**.** A random variable (RV) X is a measurable function from a sample space to a domain $\mathrm{Dom}(X)$. We write $X = x$ to denote the event that X takes value x.

Definition 2.1.2 (Joint distribution)**.** For variables $X_1, \ldots, X_n$ with domains $\mathrm{Dom}(X_i)$, the joint distribution $P(X_1, \ldots, X_n)$ assigns a probability to every assignment $(x_1, \ldots, x_n) \in \prod_i \mathrm{Dom}(X_i)$. For n binary variables, this requires 2^n entries.

2.2 Axioms (Kolmogorov)

1. $P(A) \in [0, 1]$
2. $P(\text{true}) = 1$, $P(\text{false}) = 0$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

From these we derive $P(\neg A) = 1 - P(A)$ and that any marginal distribution sums to 1.

2.3 Conditional probability, product rule, chain rule

Definition 2.3.1 (Conditional probability)**.** For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Product Rule. Rearranging gives

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

Theorem 2.3.1 (Chain rule)**.** *For any ordering variables,*

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_{i-1}, \ldots, x_1).$$

*This holds **unconditionally**, i.e. no independence assumed. The decomposition is valid for any permutation, through different orderings yield different conditional factors.*

Theorem 2.3.2 (Theorem of total probability)**.** *If $\{Y_i\}$ partitions the sample space (i.e., the Y_i are mutually exclusive and exhaustive outcomes of some variable Y), then*

$$P(X) = \sum_i P(X \mid Y_i) P(Y_i).$$

*This is the same operation as **marginalisation** (summing out)L given a joint $P(X, Y)$, we recover $P(X) = \sum_y P(X, Y = y)$.*

3 Inference via the Full Joint Distribution

Given the full joint $P(X_1, \ldots, X_n)$, *any* probabilistic query can be answered by summation.

General inference formula. Let

- X = query variable,
- $\mathbf{E} = \mathbf{e}$ = evidence (observed variables and their values),
- $\mathbf{H}$ = hidden variables (everything else).

Then

$$\mathbf{P}(X \mid \mathbf{E} = \mathbf{e}) = \alpha \mathbf{P}(X, \mathbf{E} = \mathbf{e}) = \alpha \sum_{\mathbf{h}} \mathbf{P}(X, \mathbf{E} = \mathbf{e}, \mathbf{H} = \mathbf{h}),$$

where $\alpha = 1/P(\mathbf{e})$ is a **normalisation constant** ensuring the distribution sums to 1.

The normalisation trick is useful because $P(\mathbf{e})$ may be hard to compute directly, but once we have unnormalised vector over values of X, dividing by its sum yields the same result.

Example 3.0.1 (Dental domain)**.** Consider the table below.

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	**.108**	**.012**	**.072**	**.008**
¬ cavity	**.016**	**.064**	**.144**	**.576**

$$P(\text{cavaity} \mid \text{toothache}) = \frac{P(\text{cavity}, \text{toothache})}{P(\text{toothache})} = \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064} = \frac{0.12}{0.20} = 0.6.$$

For n Boolean variables this is $O(2^n)$ in both space and time — intractable beyond around 20 variables.

4 Independence and Conditional Independence

4.1 Absolute independence

Definition 4.1.1. Events A and B are **independent**, written $A \perp B$, if and only if

$$P(A \cap B) = P(A)P(B),$$

equivalently, $P(A \mid B) = P(A)$.

Independence allows the joint to factorise, reducing storage from $O(2^n)$ to additive terms. But absolute independence is **rare** in practical domains.

4.2 Conditional independence

Definition 4.2.1. A is **conditional independent** of B **given** C, written $A \perp B \mid C$, if and only if

$$P(A \mid B, C) = P(A \mid C),$$

equivalently $P(A, B \mid C) = P(A \mid C)P(B \mid C)$.

Example 4.2.1. In the dental domain, Toothache $\perp$ Catch | Cavity: once we know whether there's a cavity, the symptom and the prob-catch outcome become independent. Combining the chain rule with conditional independence:

$$P(\text{Toothache}, \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})P(\text{Catch} \mid \text{Cavity})P(\text{Cavity}).$$

Storage reduces from $2^3 = 8$ to $2 + 2 + 2 = 6$ numbers (5 independent). In general, $O(2^n) \to O(n)$ when many variables are conditionally independent given a small set.

5 Bayes' Theorem

From the symmetry $P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$.

Theorem 5.0.1 (Bayes)**.**

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)},$$

or in vector form,

$$\mathbf{P}(A \mid B) = \alpha \mathbf{P}(B \mid A)\mathbf{P}(A).$$

Why it matters. Bayes' theorem lets us invert conditional probabilities: we can know $P(\text{symptom} \mid \text{disease})$ (which is easy to measure clinically) and recover $P(\text{disease} \mid \text{symptom})$ (the diagnostic quantity of interest), provided we know the prior $P(\text{disease})$.

5.1 The Base-Rate Fallacy

Consider a test that is 99% accurate (both sensitivity and specificity) for a disease with prevalence 1/10,000. Then

$$P(d \mid t) = \frac{P(t \mid d)P(d)}{P(t \mid d)P(d) + P(t \mid \neg d)P(\neg d)} = \frac{0.99 \cdot 0.0001}{0.99 \cdot 0.0001 + 0.01 \cdot 0.9999} \approx 0.0098.$$

Despite the high accuracy, the posterior probability of disease is under 1%, because the prior is so small. The Bayes formula makes the role of the prior explicit, which is why rare diseases yield mostly false positives even with accurate tests.

6 Bayesian Networks

6.1 Definition

A **Bayesian network** (BN) is a tuple $\big(\mathcal{G}, \{P(X_i \mid \mathrm{Pa}(X_i))\}_i\big)$ where:

- $\mathcal{G}$ is a **directed acyclic graph (DAG)** with one node per RV;
- each directed edge encodes a direct probabilistic influence;
- Each node carries a **Conditional Probability Table (CPT)** $P(X_i \mid \mathrm{Pa}(X_i))$, giving the distribution of X_i for every assignment of its parents $\mathrm{Pa}(X_i)$.

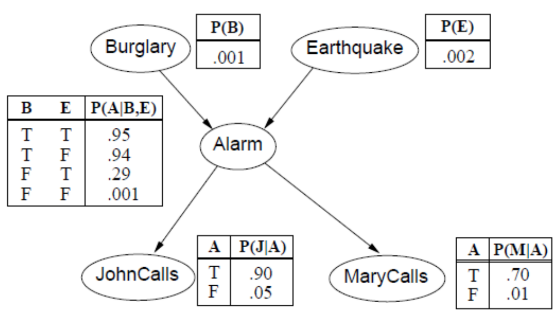

Figure 1: Burglary, Earthquake → Alarm → JohnCalls, MaryCalls

6.2 The Bayesian network assumption

Local Markov property. Each node is conditionally independent of its non-descendants given its parents:

$$X_i \perp \mathrm{NonDesc}(X_i) \mid \mathrm{Pa}(X_i).$$

6.3 Factorisation theorem

Theorem 6.3.1 (Factorisation theorem). *Under the local Markov property, the full joint distribution factorises as*

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid \mathrm{Pa}(x_i)).$$

Proof sketch. Order variables topologically (parents before children). By the chain rule,

$$P(x_1, \ldots, x_n) = \prod_{i} P(x_i \,|\, x_{i-1}, \ldots, x_1).$$

For each i, the set $\{x_{i-1}, \ldots, x_1\}$ contains only non-descendants of X_i (by topological ordering), so by the local Markov property $P(x_i \,|\, x_{i-1}, \ldots, x_1) = P(x_i \mid \mathrm{Pa}(x_i))$.

Compactness. If each node has at most k parents and variables are Boolean, the BN needs $O(2^n n)$ parameters versus $O(2^n)$ for the full joint distribution. For the burglary network shown in Figure 1:

$$1 + 1 + 4 + 2 + 2 = 10 \quad \text{vs} \quad 2^5 = 32.$$

6.4 Markov Blanket

Definition 6.4.1 (Markov blanket). The **Markov blanket** of X is the set

$$\mathrm{MB}(X) = \mathrm{Pa}(X) \cup \mathrm{Ch}(X) \cup \{\text{parents of } \mathrm{Ch}(X)\}.$$

Property. X is conditionally independent of all other nodes given $\mathrm{MB}(X)$. This is the *minimal* shielding set — useful for local computations and for Gibbs sampling.

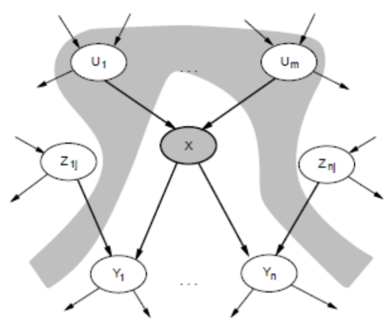

Figure 2: A node X is conditionally independent of its non-descendant nodes Z_{ij}, given its parent nodes U_i (the gray area).

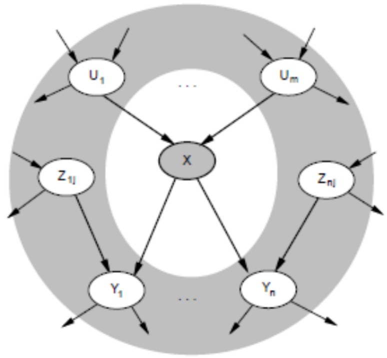

Figure 3: X is conditionally independent of all other nodes in the network, given its Markov blanket (the gray area).

6.5 Constructing a Bayesian Network

The construction algorithm enforces correctness:

1. Choose relevant variables.
2. Choose an ordering $X_1, \ldots, X_n$ (causally meaningful orderings tend to yield sparse, compact graphs).
3. For $i = 1, \ldots, n$, add X_i, set $\mathrm{Pa}(X_i)$ to a minimal subset of $\{X_1, \ldots, X_{i-1}\}$ such that $X_i \perp \{X_1, \ldots, X_{i-1}\} \setminus \mathrm{Pa}(X_i) \mid \mathrm{Pa}(X_i)$.
4. Specify each CPT $P(X_i \mid \mathrm{Pa}(X_i))$.

Note: Ordering matters substantially. A causal ordering (causes before effects) typically gives the most compact network; reverse orderings can produce dense graphs that, while still correct, defeat the purpose of using a BN.

7 Exact Inference in Bayesian Networks

The general query: given evidence $\mathbf{E} = \mathbf{e}$, compute $\mathbf{P}(X \mid \mathbf{e})$ where $\mathbf{Y}$ are the hidden variables.

$$\mathbf{P}(X \mid \mathbf{e}) = \alpha \sum_{\mathbf{y}} \mathbf{P}(X, \mathbf{e}, \mathbf{y}) = \alpha \sum_{\mathbf{y}} \prod_{i=1}^{n} P(x_i \mid \mathrm{Pa}(x_i)).$$

7.1 Inference by Enumeration

For the burglary example, computing $\mathbf{P}(B \mid j, m)$:

$$\mathbf{P}(B \mid j, m) = \alpha \mathbf{P}(B) \sum_{e} P(e) \sum_{a} P(a \mid B, e) P(j \mid a) P(m \mid a).$$

Evaluating:

$$\mathbf{P}(B \mid j, m) = \begin{bmatrix} 0.284 \\ 0.716 \end{bmatrix},$$

i.e. given both neighbours call, there's about a 28% chance of a burglary.

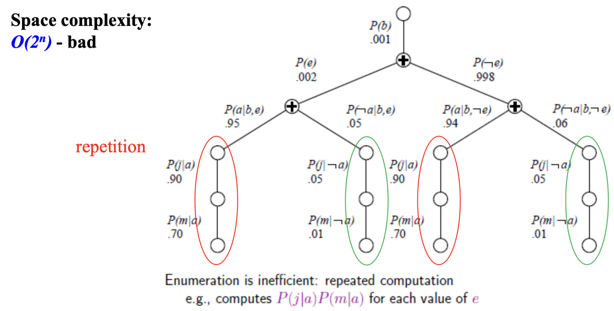

Figure 4: Enumeration tree diagram showing the repeated $P(j \mid a)P(m \mid a)$ subcomputations.

Complexity. In the worst case, $O(2^n n)$ time. The recursive enumeration tree exhibits **repeated subcomputation** of the same factors — the source of inefficiency.

7.2 Inference by Variable Elimination

Idea. Evaluate the summation right-to-left, caching intermediate results as **factors** (multidimensional arrays indexed by variable assignments).

Two factor operations

1. **Pointwise product.** For factors $f_1(\mathbf{X}_1)$ and $f_2(\mathbf{X}_2)$, the product

$$(f_1 \cdot f_2)(\mathbf{X}_1 \cup \mathbf{X}_2)$$

   assigns to each joint assignment the product of corresponding entries.

2. **Summing out (variable elimination).** Given $f(\mathbf{X}, Y)$,

$$\left(\sum_Y f\right)(\mathbf{X}) = \sum_{y \in \mathrm{Dom}(Y)} f(\mathbf{X}, y).$$

Algorithm 1 Inference by variable elimination algorithm sketch

factors $\leftarrow \emptyset$
for each $v \in$ reverse(topological order) **do**
  factors $\leftarrow$ MAKE-FACTOR$(v, e) \cup$ factors
  if v is hidden **then**
    factors $\leftarrow$ SUM-OUT$(v, \text{factors})$
  end if
end forreturn NORMALISE(POINTWISE-PRODUCT(factors))

For our query, the sequence is: form $f_4 \cdot f_5$, multiplying in f_3, sum out A to get $f_6(B, E)$, multiply by f_2, sum out E to get $f_7(B)$, multiply by f_1, normalise. Same answer $[0.284, 0.716]^\top$ but no redundant work.

Complexity. Time and space depend on the size of the largest intermediate factor, which depends on the **elimination order**. Finding the optimal order is NP-hard, but a useful heuristic is to eliminate the variable that minimises the size of the next factor produced. In the worst case (e.g., densely connected networks), variable elimination is still exponential — exact inference in BNs is #P-hard in general.

7.3 Approximate Inference

When exact inference is infeasible, sampling-based methods scale better:

- **Direct sampling** (prior sampling, rejection sampling)
- **Likelihood weighting** (sample non-evidence; weeight by evidence likelihood)
- **Gibbs sampling** (MCMC over the Markov blanket of each variable)

As the number of samples $N \to \infty$, empirical fractions converge to the true posterior. This is the route to inference in large networks.

8 Naive Bayes as a Bayesian Network

The Naive Bayes classifier corresponds to a BN with a particularly restrictive topology:

- Class variable Y as the unique root.
- Features $X_1, \ldots, X_d$ as leaves with Y as their sole parent.

The structure encodes the assumption $X_i \perp X_j \mid Y$ for all $i \neq j$, yielding

$$P(y, \mathbf{x}) = P(y) \prod_{j=1}^{d} P(x_j \mid y),$$

and the classifier

$$\hat{y}(\mathbf{x}) = \operatorname*{argmax}_y P(y) \prod_{j=1}^{d} P(x_j \mid y).$$

This is rarely a literally true assumption, but the resulting classifier is often surprisingly competitive. It is partly because for decision-making we only need the arg max, not calibrated probabilities.

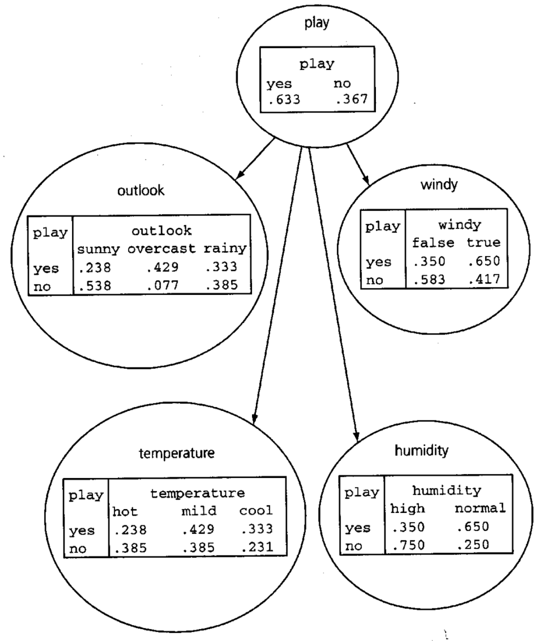

Figure 5: In the weather dataset, outlook, windy, temperature and humidity are independent given play.

9 Learning CPTs from Data

Given a dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$ with all BN variables observed, the **maximum likelihood estimate** of each CPT entry is a simple count ratio:

$$\hat{P}(X_j = x \mid \mathrm{Pa}(X_j) = \mathbf{u}) = \frac{\sum_{i=1}^{n} \mathbf{1}\{X_j^{(i)} = x \wedge \mathrm{Pa}(X_j)^{(i)} = \mathbf{u}\}}{\sum_{i=1}^{n} \mathbf{1}\{\mathrm{Pa}(X_j)^{(i)} = \mathbf{u}\}}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function.

Caveats.

- Sparse cells (rare parent configurations) give noisy or undefined estimates. **Laplace / Dirichlet smoothing** (adding pseudo-counts) is standard.
- With hidden variables or missing data, MLE has no closed form, then use Expectation-Maximisation (EM) algorithm.
- Learning the *structure* (the DAG itself) from data is a much harder problem, it is typically posed as a search over DAGs by likelihood plus a complexity penalty (BIC/MDL).